P1 Power page

3 × AA through a resettable fuse into V+ (H2) and GND (H3). Build it first.

Draft: tagged steps may change

H1 BUS A · NC

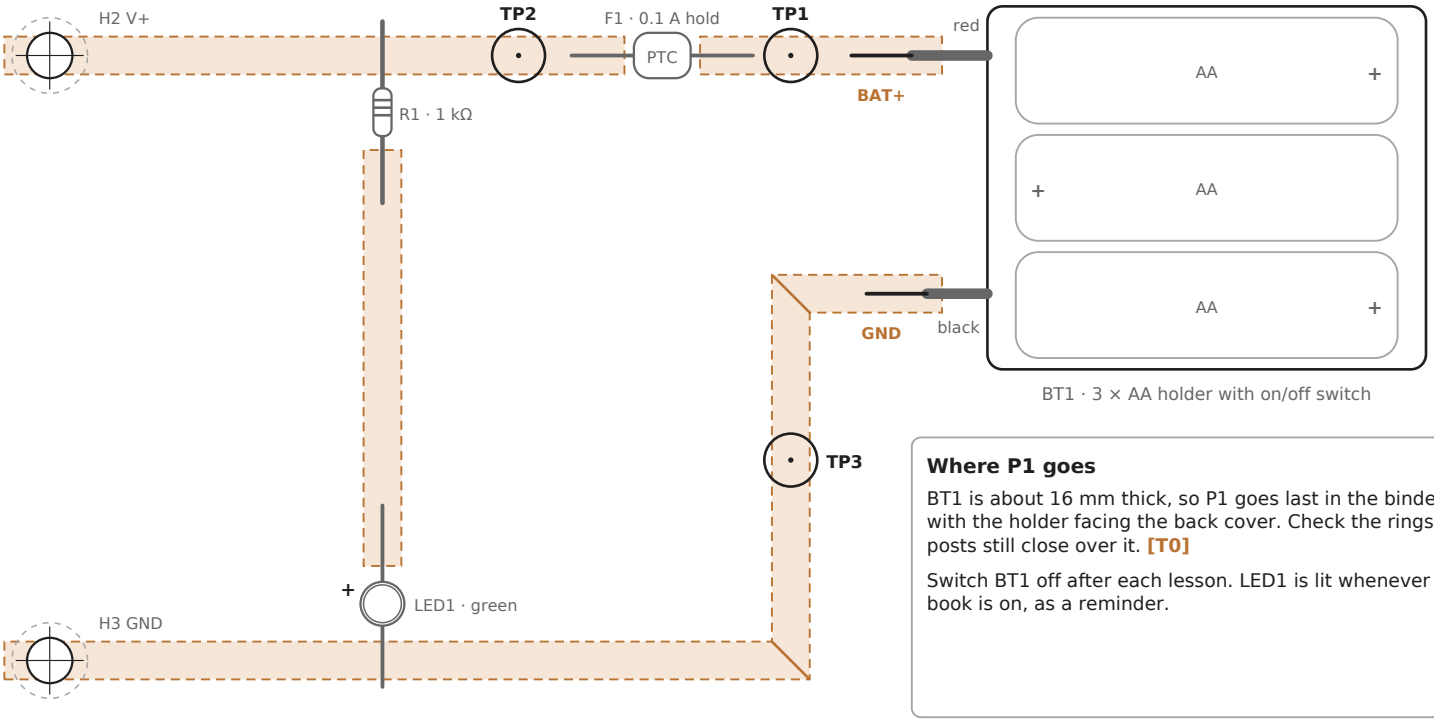
You need

- BT1: 3 × AA holder with an on/off switch and wire leads **[P]**
- 3 AA alkaline cells **[P]**
- F1: resettable fuse (PTC), 0.1 A hold **[P]**
- R1: 1 kΩ (brown-black-red) **[P]**
- LED1: green LED, 5 mm
- 5 mm copper tape, double-sided foam tape

Build it

1. Lay the four lanes, one piece of tape each. Fold at the creases; never cut.
2. Punch the holes and set eyelets in H2 and H3. **[T0]**
3. Stick BT1 down with foam tape, wires toward the spine.
4. Strip 8 mm from each wire and twist it. Lay red along BAT+ and black along GND. **[T0]**
5. Place F1 and R1 across their gaps, either way round. Place LED1 as printed: long leg on the +.
6. Hold every lead and wire end flat with a patch of copper tape. **[T0]**
7. Fit the cells, switch BT1 on and fill in Check. Then switch it off.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.



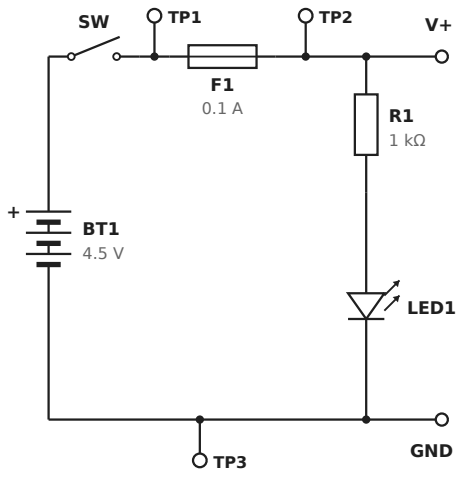
BT1 · 3 × AA holder with on/off switch

Where P1 goes

BT1 is about 16 mm thick, so P1 goes last in the binder with the holder facing the back cover. Check the rings or posts still close over it. **[T0]**

Switch BT1 off after each lesson. LED1 is lit whenever the book is on, as a reminder.

Schematic



SW is the switch built into the holder.

Check Switch BT1 on. Black probe on TP3.

TP1 _____ V **TP2** _____ V **LED1 lit?** Y / N

Fresh cells read about 4.5–4.8 V. TP1 and TP2 should match.

Explain F1 is a fuse that resets itself.

What could happen to the book if something shorted and F1 wasn't there?

Before you bind it Tick each one.

- ☐ BT1 is switched off.
- ☐ No bare wire end touches the other lane.
- ☐ P1 is the last page in the binder.

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.